

ABSTRACT

The report is a detailed description of my three week internship carried out as part of B.E in computer science & design. The internship was carried by the institution DSATM.

- ✚ This report focuses on the studies, by considering students performance in exams with the help of visualization tool **Tableau Public Software**.
- ✚ The detailed report on students performance is provided for analysis along with subject wise scores , parental details as well as reading and writing scores. I was curious to analyse the content of students data sets and create these simple, interactive and exciting visualization with **Tableau**.
- ✚ I was involved from Oct 11 to Oct 29,2022. This Internship was based on Data Analysis using Tableau.
- ✚ In this Internship I have learnt basic knowledge to use Tableau i.e.to connect the data files of different format like CSV, Excel, PDF. Furthermore, I have learnt to create the chart considering Scatterplots, Bar Charts, Line Graphs and Pie Charts they are the backbone of performing visual analysis on non-geospatial data.
- ✚ Tableau Time series Analysis lets you utilize its drag and drop feature to evaluate times by day of the week. Learned to create Dashboard which is collections of several views, letting you compare a variety of data simultaneously and creating a storyline in tableau is a sequence of visualization to convey information, use to narrate the data, demonstrate decisions related to outcomes.

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1. INTRODUCTION

1.1 Tableau

Tableau Software is an interactive data visualization software company headquartered in Seattle, Washington, United States focused on business intelligence.

- Tableau is the fast growing and powerful data visualization tool. Tableau is a business intelligence tool which helps us to analyse the raw data in the form of the visual manner; it may be a graph, report, etc.
- Tableau was founded by **Pat Hanrahan, Christian Chabot and Chris Stolte** in 2003.
□ Tableau could be better understood with help of Mastering In Tableau 10 Course.

Example: - If you have any data like **Big Data, Hadoop, SQL**, or any cloud data and if you want to analyse that given data in the form of pictorial representation of data, you can use Tableau.

The Tableau Product Suite consists of

- Tableau Desktop
- Tableau Public
- Tableau Online
- Tableau Server

1.2 Data Analysis

- Data analysis is very fast with Tableau, and the visualizations created are in the form of worksheets and dashboards. Any professional can understand the data created using Tableau.
- Tableau software doesn't require any technical or any programming skills to operate. Tableau is easy and fast for creating visual dashboards.
- Visualization is one of the best method of understanding all the behaviours of the given sets of data.
- Tableau is one of the tool used for sorting data ,segregation ,representing data as graphs or charts.

❖ Using this software, I have visualized the given data set of **Students Performance in Exams**. I visualized six worksheets and one dashboard,

categories of worksheets are:

1. Girls and boys count differentiated
2. Ethnicity based differentiation into groups
3. Parental level of education.
4. Lunch provided type
5. Reading vs Writing scores (in common and differentiated by gender)
6. Interactive worksheet with marks distribution at different levels

Analyzing the Data in a Better Way

Analyzing reports helps business stakeholders focus on the areas that require attention. The visual mediums help analysts understand the key points needed for their business. Whether it is a sales report or a marketing strategy, a visual representation of data helps companies increase their profits through better analysis and better business decisions.

Faster Decision Making

Humans process visuals better than any tedious tabular forms or reports. If the data communicates well, decisionmakers can quickly take action based on the new data insights, accelerating decision-making, and business growth simultaneously.

Making Sense of Complicated Data

Data visualization allows business users to gain insight into their vast amounts of data. It benefits them to recognize new patterns and errors in the data. Making sense of these patterns helps the users pay attention to areas that indicate red flags or progress. This process, in turn, drives the business ahead.

There are many tools used for data visualization **Tableau** is one of the Tool it is a visual analytics platform transforming the way we use data to solve problems empowering people and organizations to make the most of their data.

1.3 Advantages of Tableau

1. Quickly Create Interactive visualization

Using drag-n-drop functionalities of Tableau, the user can create a very interactive visual within minutes. The interface can handle endless variations while also limiting you from creating

charts that are against data visualization best practices. You can check out some of the amazing visuals created at the [Tableau Gallery](#).

2. Ease of Implementation

There are many different types of visualization options available in Tableau which enhance the user experience. Also, Tableau is very easy to learn compared to Python, Business Objects and Domo, anyone without having knowledge of coding can easily learn Tableau.

3. Tableau can handle large amounts of data:

Tableau can handle millions of rows of data with ease. Different types of visualization can be created with a large amount of data without impacting the performance of the dashboards. Also, there is an option in Tableau where the user can make “live” connections to different data sources like SQL etc.

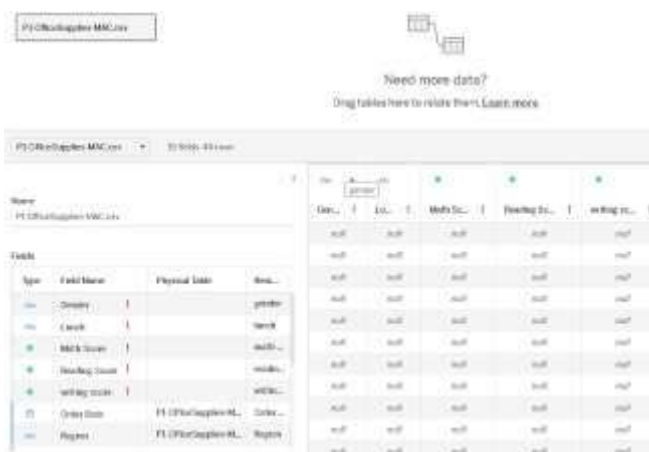


Fig:1.1

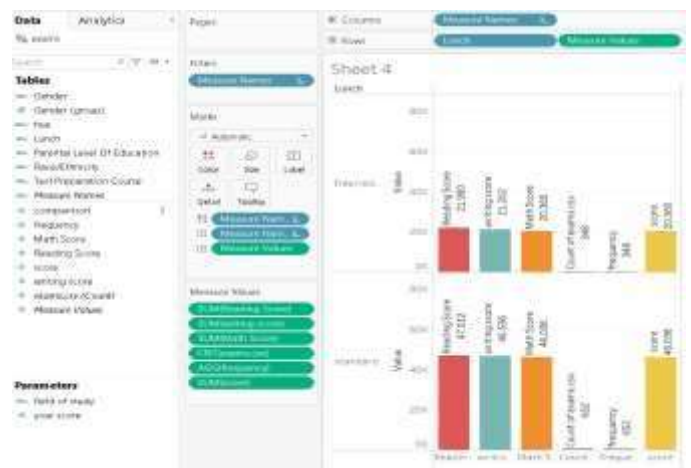


fig:1.2

4. Integrated Algebraic Operations: In Tableau we can perform algebraic operations by setting calculated fields with that of the data sets available or setting parameters etc.

5. Friendly Dashboard: tableau provides the user most compatible dashboard for best visualization where user can use worksheets created on data in dashboard or offline pictures or online weblink or even text etc.



Fig:1.3

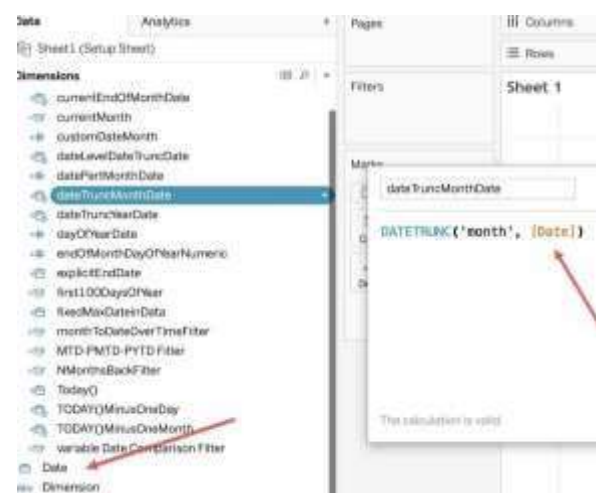


fig:1.4



Fig:1.5

6. Device compatibility: the visualizations created in tableau could be viewed in any device easily only once its published its ready to view.

2. LEARNING OUTCOME

Internship was carried out for three weeks from 11-10-22 to 29-10-22. During my internship experience, I was able to develop my skills in new fields i.e. Data visualization using Tableau software.

2.1. Week 1

In the 1st week we started our internship by understanding what tableau is, importance of Data visualization in industries and installation of tableau public in laptop.

- Tableau supports connecting to a wide variety of data, stored in a variety of places. For example, your data might be stored on your computer in a spreadsheet or a text file, or in a big data, relational etc.
- **Charts in tableau:** charts play a significant role in data visualization, The “Show Me” feature is extremely helpful since there are different types of charts available Once you drag in or double click on the dimensions and measures that you want in your analysis, you can use the Show Me feature to see the available charts that you can create by simply clicking on the chart types.
- After learning basics of Tableau, I created a dashboard with respect to case study on Apple stock history, I downloaded the data set of apple historical data from Yahoo finance and started by creating worksheets, comparing values across different dimensions and measures and bringing everything in a dashboard.

2.2. Week 2

- In 2nd week I learnt Steps to carry out **Data joining** and different types of Joins in tableau.

- **Table calculation:** Transforming values to ranking, running totals and percent of totals.
- **Data blending:** combining data from multiple sources/datasets.
- Adding **colours**, attach **labels** to tableau content and change and modify the appearance using **formatting**.
- **Data aggregation:** Gathering data and expressing in a statistical and summary form.
 - **Filters:** Minimizing size of data, removing irrelevant dimensions etc
 - **Creating maps:** Analyse and display geographically related data and present in form of maps.
 - **Working with hierarchies:** arranging entities and presenting at different levels.

2.3. Week 3 (24-10-22 to 29-10-22) ○ **Creating a storyline:** Learnt to create sequence of visualization, data narrative, provide context and demonstrate how decisions relate to outcomes.

- **Creating Dashboard:** Comparing variety of data simultaneously from the worksheet.
- **Project work:** case study on Netflix originals creating four different worksheet and bringing every worksheet together in a Dashboard for Data visualization.

2.4 Data Operations:

2.4.1 Data Blending: To prepare a dashboard data is needed so data must be

connected to Tableau. On the left side of Tableau Desktop, you will the option to connect your data from different sources. Tableau can connect with the local file and the server. After connecting to a data source connection data has to be exported as an alternate route which permits rapid connections without opening the connection dialog box.

On a single worksheet when data is Tableau blended from multiple data sources and is attached on common dimensions that is **Data blending** in Tableau. It does not create row-level joins. It should be utilized when you have related data in many data sources that you want to examine together in a single view. To combine the data blending tableau, you should first add one of the common dimensions from the primary data source to the view.

2.4.2 Data joining:

It is a very common requirement in any data analysis. You may need to join data from multiple sources or join data from different tables in a single source. Tableau provides the feature to join the table by using the data pane available under Edit Data Source in the Data menu.

2.4.3 Calculated fields:

It allows to create new data from data that already exists in the data source. When a calculation field is created, eventually a new field is created in the data source, the values or members of which are determined by a calculation. This new calculated field is saved to the data source in Tableau, and can be used to create more robust visualizations.

There are three types of Calculation field **Basic calculations** allow you to transform values or members at the data source level of detail or at the visualization level of detail, **Level of Detail (LOD)** calculations allow you to compute values at the data source level and the visualization level, **Table calculations** allow you to transform values at the level of detail of the visualization only.

2.4.4 Charts and Graphs:

The types of Chart and Graphs are

Bar Chart can be used to quickly compare data across categories, highlight differences, show trends and outliers, and reveal historical highs and lows at a glance.

Line Chart, or line graph, connects several distinct data points, presenting them as one continuous evolution. Use line charts to view trends in data, usually over time.

Pie Chart are powerful for adding detail to other visualizations. Alone, a pie chart doesn't give the viewer a way to quickly and accurately compare information.

Maps are a no-brainer for visualizing any kind of location information, whether it's postal codes, state abbreviations, country names. If you have geographic information associated with your data, maps are a simple and compelling way to show how location correlates with trends in your data.

Density Maps reveal patterns or relative concentrations that might otherwise be hidden due to an overlapping mark on a map helping you identify locations with greater or fewer numbers of data points. **Scatter Plot** are an effective way to investigate the relationship between different variables. A scatter plot presents lots of distinct data points on a single chart. The chart can then be enhanced with analytics like cluster analysis or trend lines.

Bubble Chart aren't technically their own type of visualization, Varying the size and color of circles creates visually compelling charts that present large volumes of data at once.

Tree map relate different segments of your data to the whole. As the name of the chart suggests, each rectangle in a tree map is subdivided into smaller rectangles, or sub-branches, based on its proportion to the whole.

Cluster analysis partitions the marks in the view into clusters, where the marks within each cluster are more similar to one another than they are to marks in other clusters. Tableau distinguishes clusters using color.

2.4.5 Dashboard: A **dashboard** is a consolidated display of many worksheets and related information in a single place. It is used to compare and monitor a variety of data simultaneously. The different data views are displayed all at once. Dashboards are shown as tabs at the bottom of the workbook and they usually get updated with the most recent data from the data source. While creating a dashboard, text areas, webpages and images can be added.

3. INTERNSHIP SCHEDULE

Week	Day	Tasks
1	11-10-2022	Overview, Introduction to Tableau, Installation, Trivial Exercises
	12-10-2022	Connecting Tableau to data file: CSV, Excel, PDF Case-study
	13-10-2022	Charts in Tableau
	14-10-2022	A case-study: Simple Visualization
	15-10-2022	Recap, Weekly Report Submission
2	17-10-2022	Joining data, Table Calculation
	18-10-2022	Data Blending, Adding colors, Labels, and formatting
	19-10-2022	Data Aggregation and Filters
	20-10-2022	Creating maps, working with hierarchies
	21-10-2022	Working with time and Series
	22-10-2022	Recap, Weekly Report Submission
3	24-10-2022	Creating a storyline
	25-10-2022	Creating dashboard
	26-10-2022	Project Work (on Case Study)
	27-10-2022	Report Preparation
	28-10-2022	Report Preparation
	29-10-2022	Report Submission

4. TOOL USED: TABLEAU PUBLIC

Tableau is a powerful and fastest growing data visualization tool used in the Business Intelligence Industry. It helps in simplifying raw data in a very easily understandable format.

Tableau helps create the data that can be understood by professionals at any level in an organization. It also allows non-technical users to create customized dashboard

Data analysis is very fast with Tableau tool and the visualizations created are in the form of dashboards and worksheets.

The best features of Tableau software are

- Data Blending
- Real time analysis
- Collaboration of data
- Unions

The great thing about Tableau software is that it doesn't require any technical or any kind of programming skills to operate. The tool has garnered interest among the people from all sectors such as business, researchers, different industries, etc.

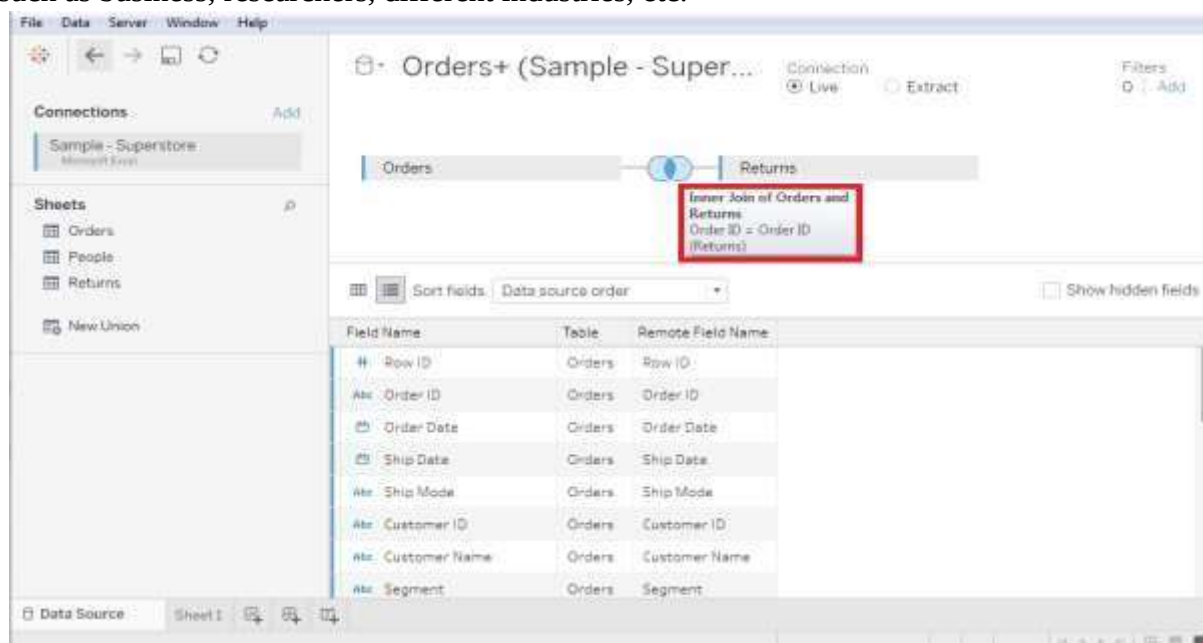
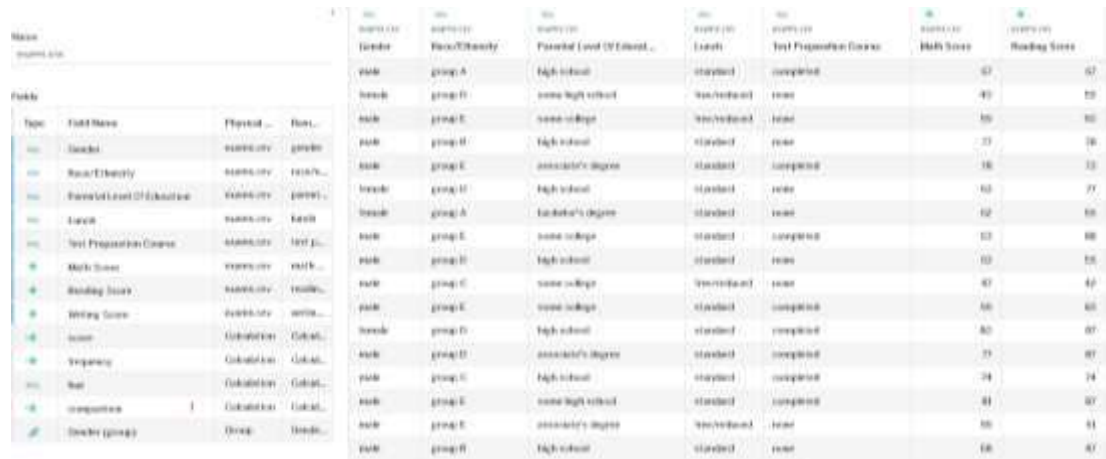


Fig:4.1

5. DATASETS USED

Tableau data sets are heterogeneous data that are sourced from various data sources and used for data visualizations. Dataset is the core component that is part of any business process related to business intelligence.



The screenshot shows a Tableau worksheet with a data table. On the left, a 'Fields' pane lists various fields categorized by type (Dimension or Measure). The main view displays a table with the following columns: Gender, Race/Ethnicity, Parental Level of Education, Lunch, Test Preparation Course, Math Score, and Reading Score. The data rows show individual student records with their respective values for these dimensions and their corresponding Math and Reading scores.

Gender	Race/Ethnicity	Parental Level of Education	Lunch	Test Preparation Course	Math Score	Reading Score
male	group A	high school	standard	completed	57	57
female	group D	some high school	free/reduced	none	49	59
male	group E	some college	free/reduced	none	59	60
male	group B	high school	standard	none	73	76
male	group E	associate's degree	standard	completed	18	12
female	group D	high school	standard	none	63	77
female	group A	associate's degree	standard	none	62	68
male	group E	some college	standard	completed	53	66
male	group D	high school	standard	none	69	78
male	group C	some college	free/reduced	none	87	87
male	group E	some college	standard	completed	66	68
female	group D	high school	standard	completed	80	87
male	group D	associate's degree	standard	completed	79	87
male	group C	high school	standard	completed	74	74
male	group E	some high school	standard	completed	81	87
male	group E	associate's degree	free/reduced	none	55	61
male	group B	high school	standard	none	68	67

Fig 5.1

DIMENSIONS : Dimensions contain qualitative values (such as names, gender, lunch, math score, reading and writing score or geographical data).

From the figure 1.1, the dimensions used are:

1. Gender
2. Parental education
3. Boys and girls number
4. Reading and writing scores
5. Food type

MEASURES: Measures contain numeric, quantitative values that you can measure.

From the figure 1.1, the measures used are:

1. Differentiation no of girls and boys
2. Calculated fields on free/standardised food
3. Setting parameters of field of study

6.DESIGN WORKSHEETS

Follow the following steps for creation of worksheets:

Steps:

1. Extraction of data: drag and drop the data sets. As shown in fig: ,and create the required joins and unions etc.
 2. Creation of calculated field : create the required calculated fields for the set of data by giving conditions as shown in fig:
 3. creation of parameters: create parameters of field as shown in fig : , Here we are creating parameters like :-your score and field of study.
 4. presentation of work: we can create visualizations with the help of calculated fields we created above and parameters .
 - We are just using drag and drop feature here in the place of rows and another parameter in columns.
 - We are setting few of the data sets on filter field in order to represent them.
 - Here we are putting fields in display unit in order to get values in the chart obtained. □
We can select type of chart we need for data representation.
 4. Once all the worksheets are prepared put them all together in one or more than on dashboard.
- After creation of dashboard do publish it in tableau public server in order to access your workit anytime and anywhere, In any device.

7. Results and Screenshots Worksheet 1

I. Number of males and females count

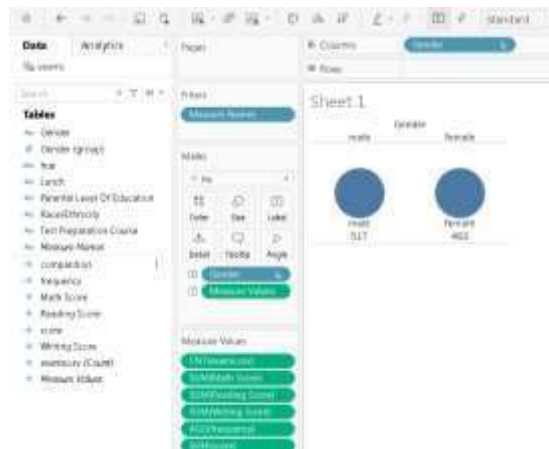


Fig: 7.2

Figure 1.3 shows the worksheet of gender count of children pie chart is used to visualize the data

Worksheet 2

II. Reading score vs Writing scores comparison:



A



B

In Fig: 7.3 A) reading vs writing scores in common B) reading vs writing scores separated for boys and girls

□ Bar chart is used for data visualization.

Worksheet 3

III. Creation of interactive worksheet:

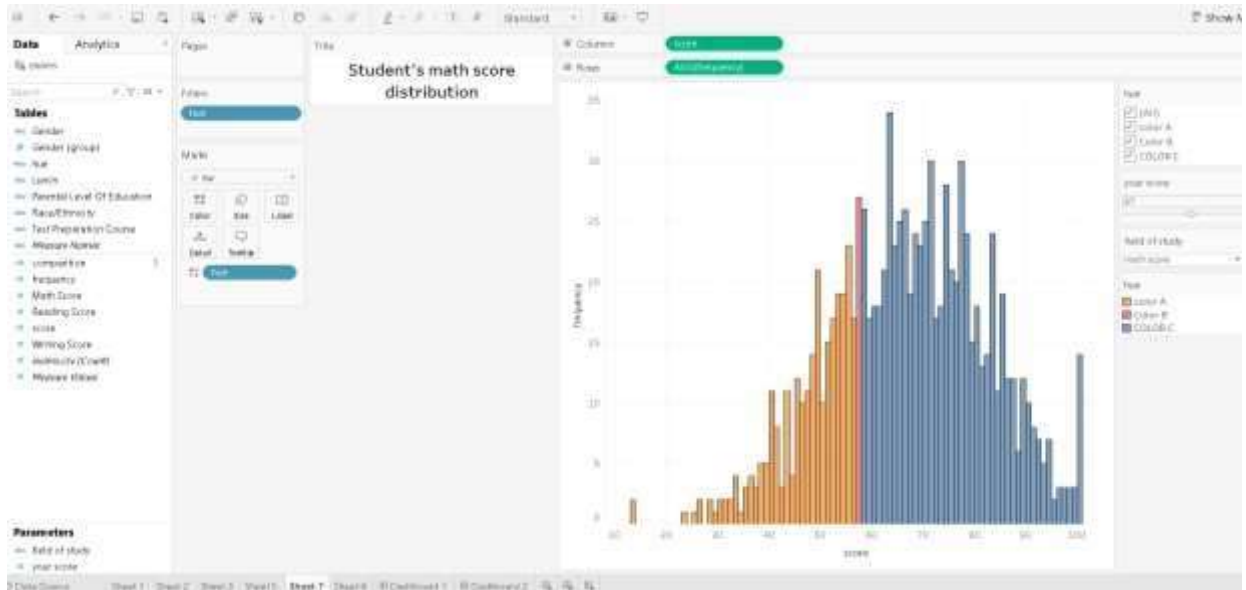


Fig: 7.4

Here we are setting two parameters or frequencies for to obtain this filter of AGG in order to visualize as a graph

We take score in columns and AGG in rows and HUE is taken common for filter as well as measure values.

Frequency –AGG is obtained by setting parameter

We are assigning three colours for you score and other two cases less than and greater then

.

Worksheet 4

IV. Performance comparison of students with different lunch structure:



Fig:7.5

In this final worksheet I have visualized performance comparison

- On observation we can conclude that the students with fee lunch performed well in exams .
- Where as students with standasdised food have taken more exams and their strength is also high.

7. DESIGN OF DASHBOARD

This Project focuses on Students performance analysis in exams, the Data analysis and statistics of marks using **tableau public software**. This study provides valuable information regarding the historical data of Netflix using the given datasets.

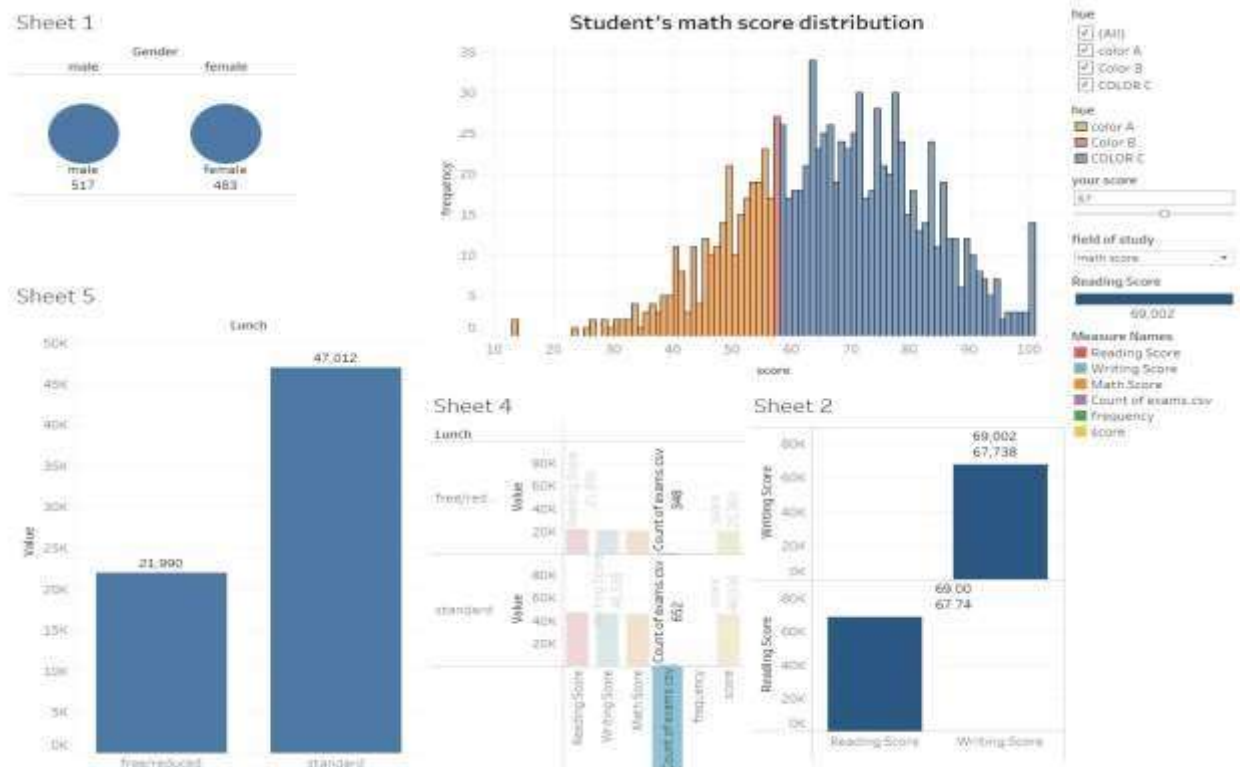


Fig: 7.1

A **dashboard** is a collection of several views, letting you compare a variety of data simultaneously.

This dashboard consists of five worksheets, the five worksheets are:

1. Number of male and female count in students
2. Overall reading vs writing scores
3. Comparison of performance of free lunch standardised students performance
4. Comparison of parents education level
5. Interactive sheet with user input score which shows your score and number of people above user and below user with color differentiation .

8. CONCLUSION

- ✚ During my internship in the college, I was able to upskill myself in a new domain i.e., data visualization using Tableau tool I was able to put these skills in practice during project work. The work was challenging, and I learned about the work and gained more confidence in myself. the overall experience was positive, and everything I learned would be useful in my future career in this field.
- ✚ Tableau is one of the software which helps in getting collaboratively result of large amount of data and get accurate results.
- ✚ In the course period I learnt how to forecast any data based on the previous state and make accurate guess, the different graphical representations like bar chart scatter chart shall allow the user to identify the density of data fields provided in order to identify highest grating .

References

- [1] Udemy Course: Tableau Hands-on Tableau Training for Data Science
- [2] Infosys Springboard: Mastering Tableau 10
- [3] Infosys Springboard: Data Visualization with Tableau
- [4] Infosys Springboard: Creating Dashboards and Storytelling with Tableau
- [5] Dataset: <https://www.superdatascience.com/>
- [6] Tableau Products - <https://www.tableau.com/products> [7] Tableau in Seven steps.

Visit the site mentioned below to access the dashboard created for the given data :

https://public.tableau.com/app/profile/tanush.reddy.k/viz/Book5_16669386518020/Dashboard1?publish=yes

